

Name: _____

SOLUTIONS

Date: _____



Year 10 Mathematics

Test 7, 2015

Topics – Probability

 60

= _____ %

Total Time: **60** minutesTotal Reading: **2** minutesTotal Working: **62** minutes

CALCULATOR FREE

Time: 10 minutes	Marks for Section: 10
	Equipment Allowed: $\frac{1}{2}$ page notes (A4 one side), Scientific calculator

1. [2 marks: 1, 1]

A bag contains 30 balls. 8 are pink, 6 are blue and the rest are white. If one ball is chosen at random from bag, find the probability that it is:

a) White

b) Pink or white

$$\frac{16}{30}$$

$$\frac{24}{30}$$

2. [1 mark]

If the probability of an event occurring is 0.35, what is the probability of it not occurring?

$$0.65$$

3. [1 mark]

There are 120 teachers at a maths conference. If one of these is chosen at random the probability that they are a Year Ten teacher is $\frac{2}{5}$. How many of the teachers are Year Ten teachers?

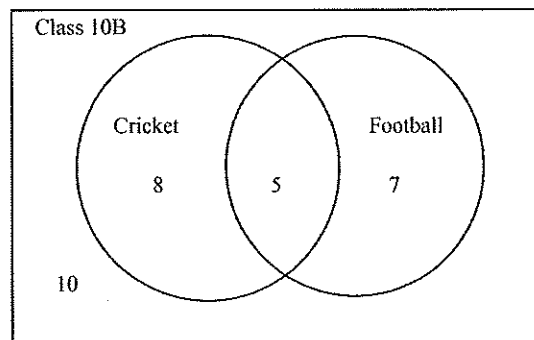
$$\frac{2}{5} \times \frac{120}{1} = \frac{240}{5}$$

$$\begin{array}{r} 48 \\ 5 \overline{) 240} \end{array}$$

$\therefore 48$ are year 10 teachers.

4. [2 marks: 1, 1]

The Venn diagram shown below shows the numbers of students in Class 10B who played cricket and football.



- a) A student from Class 10B is chosen at random. What is the probability that the student plays both football and cricket?

$$\frac{5}{30}$$



- b) A student from Class 10B is chosen at random. What is the probability that the student does not play cricket?

$$\frac{17}{30}$$



5. [4 marks: 1, 1, 2]

A single element is chosen at random from the set S: $S = \{b, c, d, 17, 19, 26, 29, *, \}\}$

What is the probability that it will be:

- a) An asterisk?

$$\frac{1}{9}$$



- b) Neither a letter or a number?

$$\frac{2}{9}$$



- c) Would there be a greater chance of it being a number or a letter? Support your answer by calculating probabilities.

$$P(\text{number}) = \frac{4}{9}$$

$$P(\text{letter}) = \frac{3}{9}$$

Greater chance of
it being a number



~ END OF SECTION ONE ~

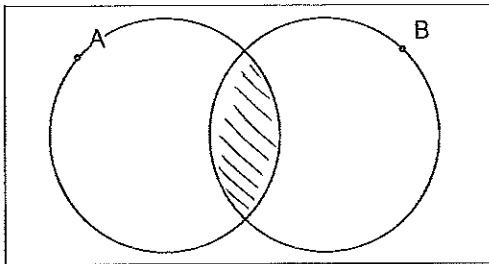
CALCULATOR ASSUMED

Time:	52 minutes	Marks for Section:	50
Reading:	2 minutes	Equipment Allowed:	½ page notes (A4 one side), Scientific calculator
Working:	50 minutes		

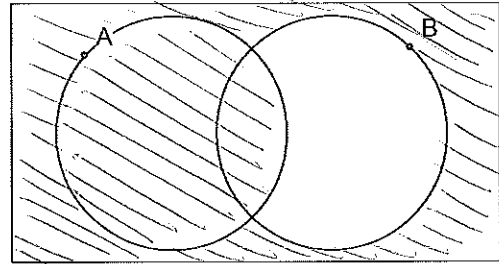
6. [2 marks: 1, 1]

In the Venn diagrams below, shade the region stated:

a) $A \cap B$



b) $A \cup \bar{B}$



7. [5 marks: 3, 2]

Given that $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$,

A = factors of 10,

B = even numbers and

C = multiple of 3.

a) List the elements in the sets below:

$$A = \{1, 2, 5, 10\}$$



$$B = \{2, 4, 6, 8, 10\}$$



$$C = \{3, 6, 9\}$$



⊖ if
no curly
brackets

b) i) Find $n(A \cup B)$

7



ii) List $B \cap C$

{6}



8. [6 marks: 2, 1, 1, 1, 1]

A sample of 120 students are asked if they eat breakfast in the mornings.

The results are shown in the table.

	Male	Female	Total
Eat Breakfast	24	31	55
Don't eat breakfast	28	37	65
Total	52	68	120

-1/2 each mistake

a) Complete the missing information in the table.

b) A person is chosen at random from the sample, what is the probability that they eat breakfast?

$$\frac{55}{120} \quad \checkmark$$

c) What is the probability that a person chosen at random is a female who eats breakfast?

$$\frac{31}{120} \quad \checkmark$$

d) If a person who prefers to eat breakfast is chosen at random. What is the probability that they are male?

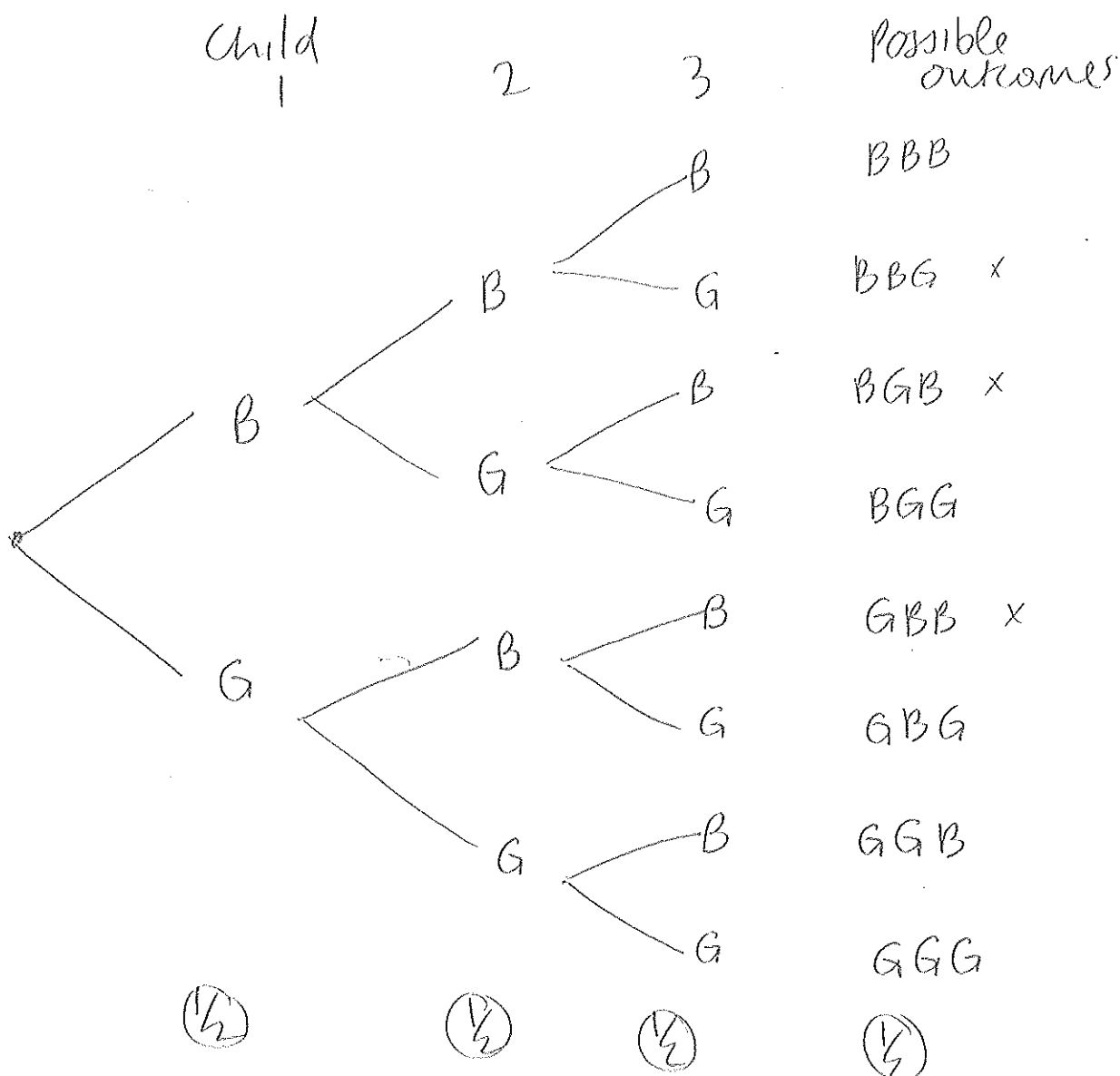
$$\frac{24}{55} \quad \checkmark$$

e) A male is chosen at random. What is the probability that he does not eat breakfast?

$$\frac{28}{52} \quad \checkmark$$

9. [5 marks: 2, 1, 2]

- a) Draw a tree diagram to represent the possible outcomes for a couple if they choose to have three children.



- b) Are these events independent?

yes ✓

- c) Use the diagram to determine the probability that the couple will have:

- i) Three girls.

$\frac{1}{8}$ ✓

- ii) Two boys and a girl in any order.

$\frac{3}{8}$ ✓

10. [5 marks: 1, 1, 1, 1, 1]

The contents of 100 boxes of smarties were counted and the results recorded.

Number of smarties	18	19	20	21	22	23	24	25
Number of boxes	9	13	x 16	18	20	15	6	3

a) Find the value for 'x' in the above table.

$$16 \quad \checkmark$$

b) What is the probability that a box chosen at random will contain exactly 22 smarties?

$$\frac{20}{100} \quad \checkmark$$

c) What is the probability that any given box will contain more than 22 smarties?

$$15 + 6 + 3$$

$$\frac{24}{100} \quad \checkmark$$

d) "It is almost certain that a box will contain less than 25 smarties." Is this statement true or false. Give evidence for your answer.

$\frac{97}{100}$ boxes $\therefore$ yes it is almost certain

$\checkmark$

e) If there were 400 boxes of smarties how many of these boxes would you expect to contain less than 20 smarties?

$$< 20 \text{ smarties} \quad 9 + 13 = 22$$

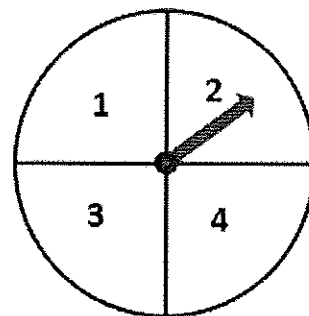
$$\therefore \frac{22}{100} = \frac{88}{400} \quad \checkmark$$

$\therefore$ you would expect 88 boxes to contain < 20 smarties.

11. [3 marks: 1, 1, 1]

The spinner on the right is spun and a coin is tossed.

a) Draw a two way table to represent the sample space.



	1	2	3	4
H	H1	H2	H3	H4
T	T1	T2	T3	T4

b) Determine the probability of obtaining a Head and a 4.

$$\frac{1}{8}$$

c) Determine the probability of obtaining a Tail or an even number.

$$\frac{6}{8}$$

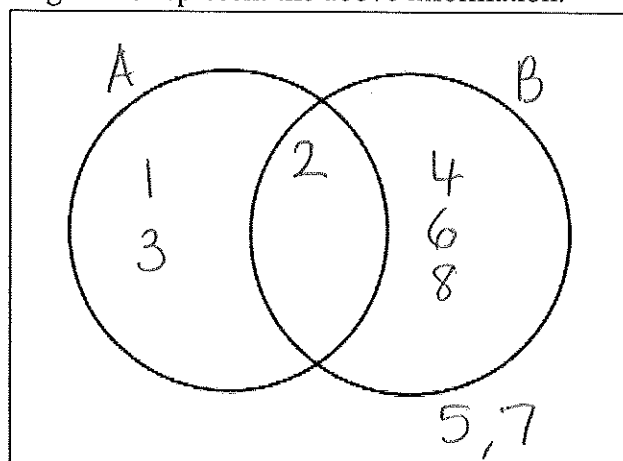
12. [4 marks: 2, 1, 1]

An eight-sided die (numbered 1-8) is rolled.

$A = \{\text{numbers less than 4}\}$

$B = \{\text{numbers that are multiples of 2}\}$

a) Draw a Venn diagram to represent the above information.



$-\frac{1}{2}$ each
mistake

b) Calculate the probability that the number is less than 4, **given** that it is a multiple of 2.

$$\frac{1}{4}$$

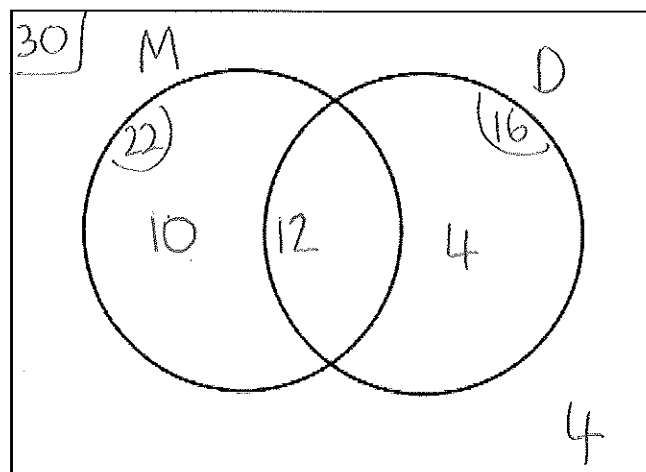
c) Determine the probability that the number is a multiple of 2, **given** it is less than 4.

$$\frac{1}{3}$$

13. [5 marks: 2, 1, 1, 1]

Miss Christie surveys her class about what chocolate they like. Out of 30 students, 16 students like dark chocolate, 22 students like milk chocolate and 4 students like neither milk nor dark chocolate.

a) Draw a Venn diagram to represent the above information.



$-\frac{1}{2}$ each mistake

b) Determine the probability that a randomly selected student like both milk and dark chocolate.

$$\frac{12}{30} \quad \checkmark$$

c) Determine the probability that a randomly selected student likes milk chocolate only.

$$\frac{10}{30} \quad \checkmark$$

d) Given a student likes dark chocolate, what is the probability they like milk chocolate.

$$\frac{12}{16} \quad \checkmark$$

14. [3 marks: 1, 1, 1]

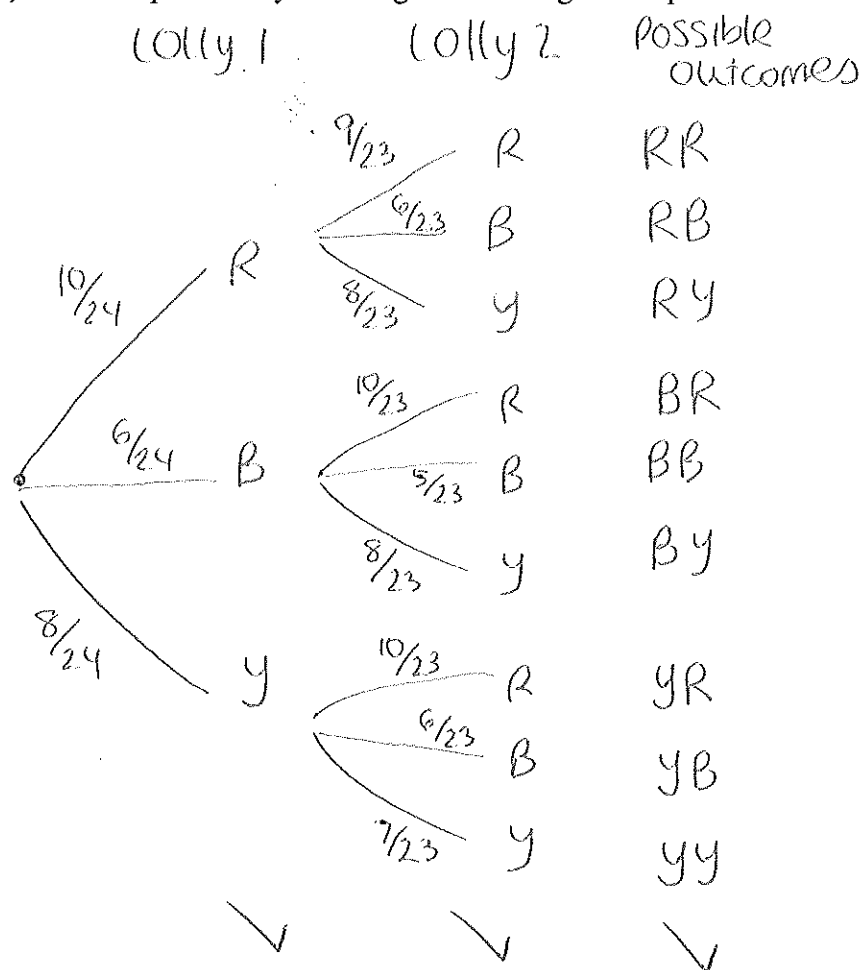
For each pair of events when rolling a regular die, decide if they are mutually exclusive or not.

Pair of events:	Mutually Exclusive
a) Rolling a number less than 3 and rolling a 6	(Yes) or No ✓
b) Rolling an even number and rolling a number greater than 4	Yes or (No) ✓
c) Rolling an odd number and rolling an even number	(Yes) or No ✓

15. [7 marks: 3, 1, 1, 1, 1]

A bag contains 24 lollies: 10 are red, 6 are black and 8 are yellow. I eat two lollies from the bag.

a) Create a probability tree diagram showing all the possible outcomes



⊖ probabilities

b) Find the probability they are both black

$$BB = \frac{6}{24} \times \frac{5}{23} = \frac{30}{552} = \frac{5}{92}$$

c) Find the probability the first is red and the second is black

$$RB = \frac{10}{24} \times \frac{6}{23} = \frac{60}{552} = \frac{5}{46}$$

d) Find the probability that given the second one is black, the first one is red

$$\frac{\frac{10}{24}}{\frac{6}{23} + \frac{5}{23} + \frac{6}{23}} = \frac{115}{204}$$

e) Find the probability neither of them is red

$$BB + BY + YB + YY$$

$$\left(\frac{6}{24} \times \frac{5}{23}\right) + \left(\frac{6}{24} \times \frac{8}{23}\right) + \left(\frac{8}{24} \times \frac{6}{23}\right) + \left(\frac{8}{24} \times \frac{7}{23}\right) = \frac{91}{276}$$

16. [5 marks: 1, 1, 1, 1, 1]

Using set notation:

If $P(A) = 0.8$, $P(B) = 0.7$ and $P(A \cap B) = 0.4$

a) Find $P(A|B)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.4}{0.7} = \frac{4}{7} \quad \checkmark$$

If $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cup B) = 0.6$

b) Find $P(A \cap B)$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ 0.6 &= 0.4 + 0.3 - P(A \cap B) \\ 0.6 &= 0.7 - P(A \cap B) \\ 0.1 &= P(A \cap B) \quad \checkmark \end{aligned}$$

If $P(A \cup B) = 0.55$, $P(A) = 0.4$ and $P(A \cap B) = 0.1$

c) Find $P(B)$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ 0.55 &= 0.4 + P(B) - 0.1 \\ 0.55 &= 0.3 + P(B) \\ 0.25 &= P(B) \quad \checkmark \end{aligned}$$

d) $P(A|B)$

$$\begin{aligned} P(A|B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{0.1}{0.25} \\ &= \frac{10}{25} \quad \checkmark \end{aligned}$$

If $P(A) = \frac{1}{2}$, $P(B) = \frac{2}{3}$ and $P(A \cap B) = \frac{5}{6}$

e) Are events A and B independent? Explain your answer.

Not independent as $P(A) \times P(B) \neq P(A \cap B)$

$\checkmark$